

ECG Signal Preprocessing Report

Offline Day 1

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Abstract

This report documents the preprocessing pipeline developed for the MIT-BIH Arrhythmia Database. The work includes exploratory analysis, frequency-domain inspection, comparative evaluation of multiple filtering techniques, beat-level segmentation, z-score normalization, and patient-level train/test splitting. Among the evaluated filters, the Stationary Wavelet Transform (SWT) achieved the highest signal fidelity with a correlation of 0.9989 and a PRD of 2.26%, consistent with findings in recent literature [1]. The final dataset contains 105,039 labeled beats across five AAMI classes, ready for downstream classification tasks.

Contents

1	Introduction	3
2	Dataset: MIT-BIH Arrhythmia Database	3
3	ECG Signal Components	3
4	Noise Sources in ECG Signals	4
5	Frequency Analysis	4
6	Filtering Techniques	4
6.1	High-Pass Butterworth Filter	5
6.2	Notch Filter	5
6.3	Moving Average Filter	5
6.4	Savitzky-Golay (SGS) Filter	5
6.5	Chebyshev Type II Bandpass Filter	5
6.6	Stationary Wavelet Transform (SWT)	5
6.7	Empirical Mode Decomposition (EMD)	6
6.8	Comparative Results	6
7	Full Pipeline	7
8	Beat Segmentation	7
9	Normalization	7

10 Train/Test Split	7
11 Summary	8

1 Introduction

Electrocardiogram (ECG) signals are fundamental diagnostic tools for monitoring cardiac health. However, raw ECG recordings are typically contaminated by various noise sources, including baseline wander, power-line interference (PLI), muscle artifacts, and electrode contact noise. Effective preprocessing is essential to ensure that subsequent analysis—whether manual interpretation or automated classification—operates on clean, reliable data.

This report presents the preprocessing pipeline applied to the MIT-BIH Arrhythmia Database, covering filtering, segmentation, normalization, and dataset splitting.

2 Dataset: MIT-BIH Arrhythmia Database

The MIT-BIH Arrhythmia Database is a benchmark dataset widely used in ECG research. Table 1 summarizes its key characteristics.

Table 1: MIT-BIH Arrhythmia Database characteristics.

Property	Value
Number of records	48
Number of unique subjects	47 (records 201 and 202 are from the same patient)
Channels per record	2 (typically MLII + V1/V2/V4/V5)
Sampling frequency	360 Hz
ADC resolution	11-bit over 10 mV range
Record duration	~30 minutes each
Total samples per channel	650,000
Total annotated beats	~110,000
Arrhythmia types	~15 classes

The MLII (Modified Limb Lead II) channel was selected as the primary signal for analysis, as it provides the clearest P-QRS-T morphology and is the standard choice in the literature.

3 ECG Signal Components

The ECG waveform consists of several key components corresponding to distinct phases of the cardiac cycle:

- **P wave** (0.67–5 Hz): Represents atrial depolarization (contraction).
- **QRS complex** (10–50 Hz): Represents ventricular depolarization. The dominant feature of the ECG, lasting 80–120 ms.
- **T wave** (1–7 Hz): Represents ventricular repolarization (electrical reset).
- **U wave**: An optional small deflection following the T wave.

The clinically relevant bandwidth of the ECG spans approximately 0.05–100 Hz, with most energy concentrated below 35 Hz.

4 Noise Sources in ECG Signals

The raw ECG signal is affected by several types of noise [1]:

- **Baseline wander** (0.1–0.5 Hz): Low-frequency drift caused by respiration, patient movement, and electrode drift.
- **Power-line interference**: Sinusoidal contamination at 50 Hz (or 60 Hz), caused by electromagnetic coupling with mains power.
- **Muscle artifacts (EMG noise)**: Broadband high-frequency noise from voluntary or involuntary muscle contractions.
- **Electrode contact noise**: Intermittent disturbances from loose electrode connections.

5 Frequency Analysis

Prior to filtering, the frequency content of the signal was examined via the Fast Fourier Transform (FFT) to identify the noise characteristics of the dataset. Analysis of Record 100 (MLII channel, first 30 seconds) revealed the following:

- The dominant signal energy resides below 20 Hz, consistent with the expected P-QRS-T frequency ranges.
- A significant DC component and low-frequency energy below 0.5 Hz was observed, confirming the presence of baseline wander.
- No prominent spike was observed at 50 or 60 Hz, suggesting that the original recording hardware applied analog filtering for PLI suppression. A minor dip around 60 Hz and a small bump near 120 Hz (second harmonic) were noted.
- The noise floor above 40 Hz ranged from -60 to -80 dB.

Basic signal statistics for the MLII channel of Record 100 are presented in Table 2.

Table 2: Signal statistics for Record 100, MLII channel.

Metric	Value
Mean	-0.3063 mV
Standard deviation	0.1932 mV
Min / Max	-2.7150 / 1.4350 mV
Range	4.1500 mV
Estimated SNR	26.64 dB

The SNR was estimated using the heuristic formula:

$$\text{SNR} = 20 \cdot \log_{10} \left(\frac{\text{Range}(x)}{\sigma_x} \right) \quad (1)$$

where $\text{Range}(x) = \max(x) - \min(x)$ approximates the signal amplitude and σ_x approximates the noise level.

6 Filtering Techniques

Six filtering techniques were implemented and evaluated on a 30-second segment of Record 100. The following subsections describe each method, and Table 3 presents a comparative

summary of all results.

6.1 High-Pass Butterworth Filter

A 4th-order Butterworth high-pass filter with a cutoff frequency of 0.5 Hz was applied using zero-phase filtering (`sosfiltfilt`) to remove baseline wander while preserving the P wave (which begins at ~ 0.67 Hz). The Butterworth design was chosen for its maximally flat passband response, ensuring minimal distortion of the retained frequency components.

6.2 Notch Filter

An IIR notch filter centered at 60 Hz with a quality factor of $Q = 30$ was applied to remove power-line interference. Since the MIT-BIH database was recorded in the United States, 60 Hz was targeted. Minimal effect was observed on this dataset due to pre-existing hardware filtering in the original recordings.

6.3 Moving Average Filter

A moving average filter with a window size of 20 samples (~ 56 ms at 360 Hz) was applied. While effective at smoothing high-frequency noise, this filter significantly degraded QRS morphology, yielding a correlation of only 0.663. This result demonstrates that the moving average filter is unsuitable for ECG applications where waveform preservation is critical.

6.4 Savitzky-Golay (SGS) Filter

The Savitzky-Golay filter fits a polynomial (order 3) to a sliding window (21 samples) rather than computing a flat average. This polynomial fitting enables the filter to follow sharp transitions such as the R peak, resulting in substantially better morphological preservation (correlation: 0.921) compared to the moving average filter.

6.5 Chebyshev Type II Bandpass Filter

A 4th-order Chebyshev Type II bandpass filter (0.5–40 Hz, 40 dB stopband attenuation) was applied. Type II was selected over Type I because its ripples occur in the stopband rather than the passband, avoiding distortion of the retained ECG components. This filter effectively combines baseline wander removal and high-frequency noise suppression in a single step.

6.6 Stationary Wavelet Transform (SWT)

The SWT decomposes the signal into approximation and detail coefficients at multiple levels without downsampling, making it shift-invariant—a key advantage over the standard Discrete Wavelet Transform (DWT). At each level j , the filters are upsampled by inserting $2^{j-1} - 1$ zeros between taps, enabling the decomposition to capture progressively lower frequency bands while maintaining the original signal length at every level.

Denoising was performed by applying soft thresholding to the detail coefficients:

$$\hat{c}_D[k] = \text{sign}(c_D[k]) \cdot \max(|c_D[k]| - \lambda, 0) \quad (2)$$

where the threshold λ was computed using the universal threshold formula:

$$\lambda = \alpha \cdot \hat{\sigma} \cdot \sqrt{2 \ln N} \quad (3)$$

with noise standard deviation estimated via the Median Absolute Deviation (MAD):

$$\hat{\sigma} = \frac{\text{median}(|c_{D,1}|)}{0.6745} \quad (4)$$

The optimal configuration, consistent with findings in [1], was:

- Wavelet: `rbio3.9` (reverse biorthogonal 3.9)
- Decomposition level: 5
- Threshold scale (α): 0.5

6.7 Empirical Mode Decomposition (EMD)

EMD is a data-driven decomposition method that extracts Intrinsic Mode Functions (IMFs) through an iterative sifting process without relying on predefined basis functions. The signal is decomposed by repeatedly identifying local extrema, constructing upper and lower envelopes via cubic spline interpolation, computing the mean envelope, and subtracting it from the signal. The process yielded 16 IMFs for the test segment.

For denoising, IMFs 2–13 were retained while the first IMF (high-frequency noise) and the last three IMFs (baseline wander and DC offset) were discarded. The reconstruction achieved a correlation of 0.970, which is reasonable but substantially lower than SWT's 0.9989.

6.8 Comparative Results

Table 3: Comparative evaluation of filtering techniques on Record 100 (30-second segment).

Filter	MSE	RMSE	PRD (%)	Correlation	SNR (dB)
High-Pass Butterworth	0.1137	0.3372	89.29	0.9753	20.17
Notch (60 Hz)	<i>Negligible change (PLI absent)</i>				
Moving Average	0.0164	0.1281	75.62	0.6631	16.93
Savitzky-Golay	0.0046	0.0678	40.05	0.9212	19.37
Chebyshev II BP	0.1185	0.3442	91.14	0.8959	19.60
SWT (rbio3.9)	0.0001	0.0085	2.26	0.9989	19.97
EMD (IMFs 2–13)	0.1183	0.3440	91.08	0.9702	20.08

The SWT filter achieved the best performance across all metrics, with a correlation of 0.9989 and a PRD of only 2.26%, confirming that it preserves the ECG morphology with minimal distortion. The high PRD values for the high-pass, Chebyshev II, and EMD filters are primarily attributable to DC offset removal rather than signal degradation, as reflected by their relatively high correlation values.

The moving average filter exhibited the lowest correlation (0.663), confirming its unsuitability for ECG preprocessing where morphological fidelity is required.

7 Full Pipeline

Based on the comparative analysis, the following filter chain was selected and applied to all 48 records (MLII channel):

$$\text{Raw ECG} \xrightarrow{\text{HP Butterworth (0.5 Hz)}} \xrightarrow{\text{Notch (60 Hz)}} \xrightarrow{\text{SWT (rbio3.9, L5, } \alpha=0.5)} \text{Clean ECG}$$

8 Beat Segmentation

Individual heartbeats were extracted using the cardiologist-annotated R-peak positions from the database annotation files. A fixed window of 250 samples (~ 694 ms at 360 Hz) centered on each R-peak was used, capturing one full P-QRS-T complex. Beats too close to signal boundaries were excluded.

Annotation symbols were mapped to the five AAMI standard classes as shown in Table 4.

Table 4: AAMI beat classification mapping.

AAMI Class	Description	MIT-BIH Symbols
N	Normal	N, L, R, e, j
S	Supraventricular ectopic	A, a, J, S
V	Ventricular ectopic	V, E
F	Fusion	F
Q	Paced / Unknown	/, f, Q

A total of 105,039 beats were extracted across all 48 records.

9 Normalization

Z-score normalization was applied independently to each beat:

$$\hat{x}[n] = \frac{x[n] - \mu}{\sigma} \quad (5)$$

where μ and σ are the mean and standard deviation of the individual beat. This ensures that each beat has zero mean and unit variance, allowing the model to focus on morphological (shape) differences rather than amplitude variations caused by inter-patient differences in electrode placement, body composition, and signal strength.

10 Train/Test Split

A patient-level split was performed to prevent data leakage. Records 201 and 202 (same patient) were assigned a shared patient identifier to ensure they remain in the same partition. `GroupShuffleSplit` from `scikit-learn` was used with the patient ID as the grouping variable.

Table 5: Train/test split summary (patient-level, zero overlap).

Split	Beats	Patients	N (%)	S (%)	V (%)	F (%)	Q (%)
Train	83,736	36	86.3	3.2	7.5	0.5	2.5
Test	21,303	9	84.8	0.3	4.6	1.7	8.5

The significant class imbalance (N class dominates at $\sim 85\%$) will be addressed at the model level through class-weighted loss functions rather than data-level resampling techniques, as synthetic oversampling methods (e.g., SMOTE) risk generating physiologically implausible ECG morphologies.

11 Summary

The complete preprocessing pipeline is summarized as follows:

1. Loaded all 48 records from the MIT-BIH Arrhythmia Database (MLII channel).
2. Applied a three-stage filter chain: High-Pass Butterworth (0.5 Hz) $\rightarrow$ Notch (60 Hz) $\rightarrow$ SWT (`rbio3.9`, level 5, threshold scale 0.5).
3. Segmented individual heartbeats using annotated R-peak positions (250-sample windows).
4. Mapped annotations to five AAMI classes (N, S, V, F, Q).
5. Applied per-beat z-score normalization.
6. Split into training (83,736 beats, 36 patients) and testing (21,303 beats, 9 patients) sets with zero patient overlap.
7. Saved the processed dataset as `mitbih_preprocessed.npz`.

The dataset is now ready for downstream classification tasks.

References

- [1] Ádám, N., Val’ko, D., Balogh, Z., Madoš, B., & Hurtuk, J. (2025). Comparative evaluation of filtration techniques for ECG signal denoising with emphasis on stationary wavelet transform. *Scientific Reports*, 15, 42514. <https://doi.org/10.1038/s41598-025-26476-1>